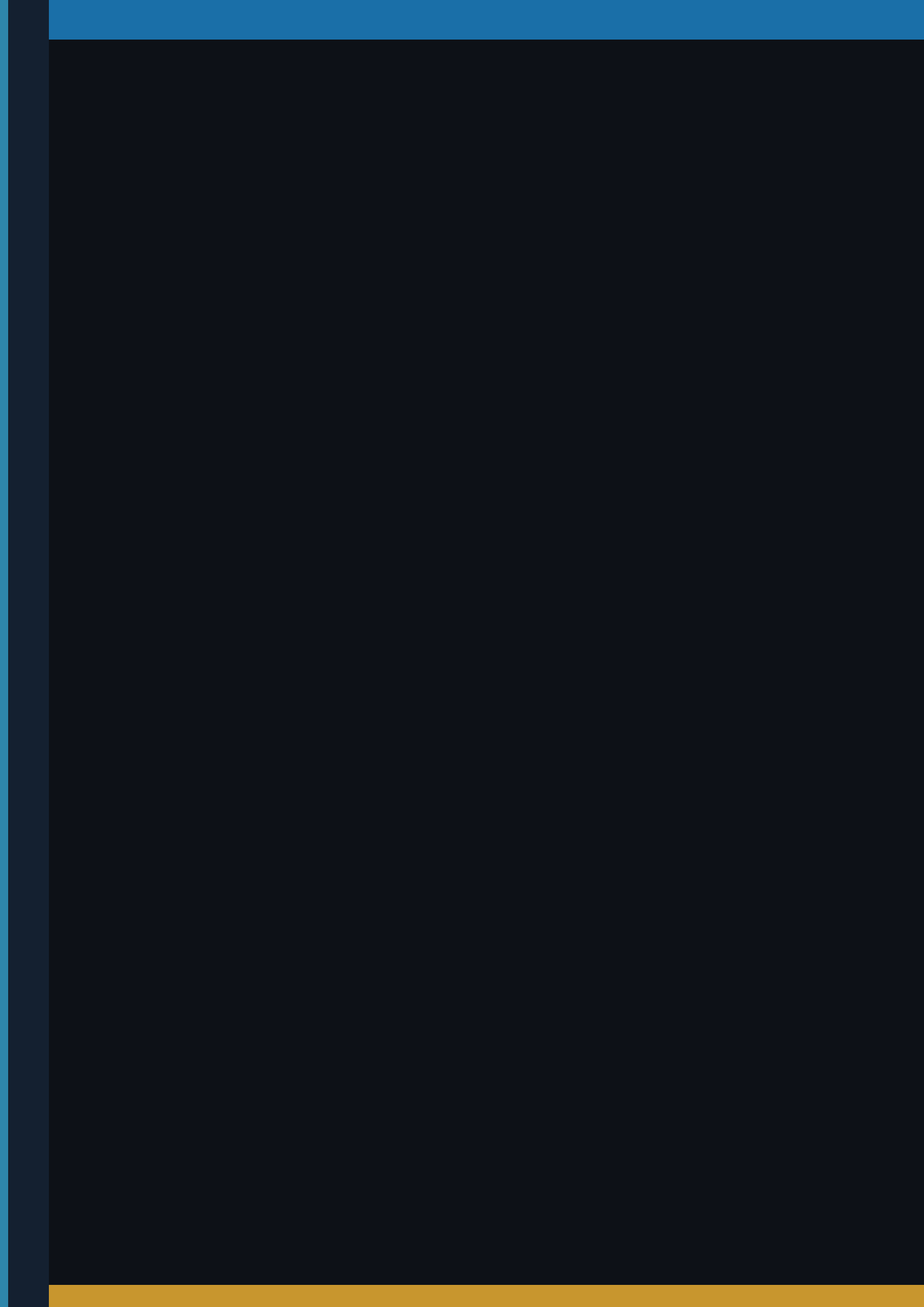


MATHEMATICS CURRICULUM FOR AI ENGINEERING 2025

Curriculum for Proof, Operators, Dynamics,
Uncertainty, and Architectural Compression

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About This Guide

This curriculum is designed for a researcher and engineer operating at the intersection of machine learning architecture, nonlinear physics, integrable systems, formal specification, and cross-domain pattern recognition. It is not a generic mathematics survey. Every phase, resource, and mastery criterion has been selected to serve four specific capabilities:

- **Formalize intuition cleanly** — convert structural insight into precise objects and theorems.
- **Separate analogy from isomorphism** — distinguish surface similarity from genuine structural equivalence.
- **Prove or falsify load-bearing claims** — bring mathematical discipline to architectural assertions.
- **Compress deep ideas into lawful machinery** — turn broad frameworks into testable, engineerable components.

The curriculum is organized into eight phases — an ordered progression from proof discipline through functional analysis and optimization — followed by a synthesis track that translates mathematical content back into architectural practice. Resources are drawn primarily from MIT OpenCourseWare, Stanford Engineering Everywhere, the MIT Underactuated Robotics notes, and OpenStax, with the awesome-math ecosystem used as a secondary discovery index

Design Principles

- Proof-capable over survey-level: every phase produces certifiable competencies.
- Engineering-relevant throughout: mathematics is traced back to computational and architectural consequence.
- Three-pass method: intuition → formal layer → engineering layer for every subject.
- Translation-first synthesis: a parallel notebook maps formal objects to system design.
- Mastery gates over completion metrics: depth beats breadth.

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Curriculum Structure and Phase Map

The curriculum is organized as a layered progression, not a random library. Phases 0–4 form the core spine and should be treated as sequential prerequisites. Phases 5–6 are advanced modules that run in parallel once the spine is stable. The synthesis track runs throughout the entire programme.

Phase	Focus	Why It Matters
0	Proofs, Discrete Math, Calculus Hygiene	Establishes rigor and symbolic discipline; prerequisite for all subsequent phases.
1	Linear Algebra — Matrix and Operator Viewpoint	Representation, operators, eigenstructure, geometry of transforms.
2	Real Analysis	Convergence, precision, proof maturity; hardens informal intuitions into certified claims.
3	Probability, Statistics, Information Theory	Uncertainty, evidence, entropy, inference; converts informal language into exact definitions.
4	Differential Equations, Dynamical Systems, Control	Regimes, stability, attractors, feedback; formalizes training dynamics and governed runtime.
5	Abstract Algebra, Differential Geometry, Complex Analysis	Symmetry, curvature, structural depth; deepens vocabulary for advanced architecture work.
6	Functional Analysis, Numerical Analysis, Optimization	Infinite-dimensional structure, computational methods, solvable formulations.
S	Synthesis Track (parallel)	Live translation of formal mathematics back into architectural objects, invariants, and experiments.

The ordering is chosen to maximize representational power early (Phase 1), proof maturity mid-spine (Phase 2), and dynamics/control intuition before the advanced modules (Phase 4). The synthesis track is not optional — it is what converts a mathematics course into an engineering capability.

2

PHA SE 0

Proof Discipline and Calculus Hygiene

Making reasoning exact before building on it

2.1 Objectives

This phase establishes the foundational discipline required for all subsequent phases. The goal is not exposure to new ideas but the hardening of existing intuition into exact, verifiable reasoning. Particular focus is placed on proof structure, formal manipulation of quantifiers, and invariant reasoning in simple systems.

- Direct proof, proof by contradiction, contrapositive, and mathematical induction
- Sets, functions, relations, and formal definitions
- Counting, recurrences, and discrete probability language
- Clean symbolic writing without handwaving
- Invariants in simple discrete systems
- Calculus repair: derivatives, integrals, single-variable mechanics (as needed)

2.2 Recommended Resources

Tier	Source	Description
Primary	MIT — Mathematics for Computer Science	Proof methods, logic, induction, state machines, graph theory, counting, recurrences, and discrete probability. Explicitly emphasizes definitions and proofs. Lecture notes, problem sets, and a full textbook are freely available on MIT OCW.
Primary	OpenStax — Calculus Volume 1	Clean, free baseline for functions, derivatives, integrals, and single-variable mechanics. Use as repair or review material for calculus hygiene.
Discovery	awesome-math	Curated index for locating additional proof primers and discrete mathematics references. Use as a secondary discovery layer only; do not use as the primary sequence.

2.3 Mastery Test

Mastery Test

- ✓ Write a clean proof by induction without handwaving, including base case, inductive step, and conclusion.
- ✓ Manipulate quantifiers carefully and convert between $\forall/\exists$ forms and their negations.
- ✓ Reason about invariants in a simple state machine: identify the invariant, prove it holds at each step.
- ✓ Derive a closed-form recurrence solution using the characteristic equation method.
- ✓ Write a proof by contradiction for a basic number-theoretic claim.

Deliverable Practice

- Maintain a proof notebook: one completed proof per study session.
- Maintain a definitions ledger: formal definitions for every new object encountered.
- Maintain an error log: record every proof failure and the precise step where the argument broke.
- The error log is where rigor is actually built — it is not optional.

3

PHA SE 1

Linear Algebra and the Operator Viewpoint

Representation, geometry, eigenstructure

3.1 Why This Phase Matters

Linear algebra is likely the highest-return phase relative to your current work. You already think in operators, transforms, latent structure, and regimes. This phase converts those intuitions into exact objects with provable properties.

After this phase, your notions of representation, compression, state, modulation, routing, and the geometry of transformation should become substantially sharper. The connection between eigenstructure and dynamical behavior is the single most important bridge between this phase and Phase 4.

3.2 Core Topics

- **Column space, null space, rank, and fundamental theorem of linear algebra**
 - Geometrically: what transformations preserve, collapse, and cannot reach.
 - Computationally: row reduction, pivot structure, solution existence and uniqueness.
- **Orthogonality, projection, and least squares**
 - Projection as the best approximation in a subspace.
 - Least squares via normal equations and the pseudo-inverse.
- **Eigenvalues, eigenvectors, and diagonalization**
 - Eigenvalues as invariant directions; characteristic polynomial.
 - Diagonalization conditions; connection to the spectral theorem.
- **Singular Value Decomposition (SVD)**
 - Geometric interpretation: rotation, scaling, rotation.
 - Low-rank approximation, conditioning, and noise filtering.
- **Quadratic forms and positive definite structure**
 - Positive definiteness as a certificate of minimum structure.
 - Connection to Lyapunov stability and optimization landscapes.
- **Linear dynamical systems**
 - State-space models: $x_{k+1} = Ax_k$.
 - Stability via eigenvalue conditions; modes and modal decomposition.
- **Controllability and observability**
 - Conceptual: can a system be driven to any state? Can its state be inferred from outputs?
 - Rank conditions for Gramians.

3.3 Recommended Resources

Tier	Source	Description
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Primary	MIT — 18.06 / 18.06SC Linear Algebra	Matrix theory and linear algebra with emphasis on topics useful across disciplines. Covers systems of equations, vector spaces, determinants, eigenvalues, similarity, and positive definite matrices. Self-study version includes lecture videos and problem sets. MIT OCW, freely available.
Primary	Stanford SEE — EE263 Introduction to Linear Dynamical Systems	Applied linear algebra connected directly to circuits, signal processing, communications, and control systems. Bridges Phase 1 and Phase 4 effectively. Full course materials available.
Companion	MIT — Matrix Calculus for Machine Learning and Beyond	Treats derivatives as linear operators; generalizes Jacobians and matrix functions. Directly bridges linear algebra to the ML-adjacent operator perspective.
Discovery	awesome-math — Linear Algebra section	Secondary index for additional texts (e.g., Axler's Linear Algebra Done Right for coordinate-free treatment, Horn and Johnson for advanced matrix analysis).

3.4 Mastery Test

Mastery Test

- ✓ Derive the least-squares solution via the normal equations and explain geometrically what projection is doing.
- ✓ Compute and interpret the SVD of a 3×3 matrix; explain the geometry of each factor.
- ✓ Given a linear map, identify its column space, null space, and rank without computation.
- ✓ Explain why a 2×2 linear system $x_{k+1} = Ax_k$ is stable if and only if all eigenvalues satisfy $|\lambda| < 1$.
- ✓ Move fluidly between coordinate-free language (linear map, kernel, image) and matrix representation.

4

PHA SE 2

Real Analysis

Hardening intuition into certified claims

4.1 Objectives

Real analysis is where intuition gets hardened. The goal is to rebuild calculus from first principles — not for ceremony but because the informal versions of continuity, convergence, and approximation that suffice for engineering practice will fail when claims become load-bearing. Every theorem in this phase should be understood as a precise statement of when something that seems obvious is actually true.

This phase will eliminate hidden assumptions, vague smoothness language, unjustified limit manipulations, and informal reasoning about stability.

4.2 Core Topics

- Real numbers and completeness — the Archimedean property, supremum, and why $\mathbb{R}$ has no gaps
- Sequences and series — convergence, Cauchy sequences, and absolute/conditional convergence
- ε - δ rigor — the formal definitions of limit and continuity and their proof patterns
- Differentiability — mean value theorem, Taylor's theorem with remainder
- Uniform convergence — why it matters for interchanging limits and the distinction from pointwise
- The Riemann integral — definition, Riemann sums, and the fundamental theorem
- Compactness — Heine-Borel, extreme value theorem, and uniform continuity on compact sets
- Fixed-point theorems — Banach contraction mapping as a precursor to Phase 6

4.3 Recommended Resources

Tier	Source	Description
Primary	MIT — Real Analysis (18.100A/B/C)	Teaches how to prove theorems in analysis and how to write proofs. Rigorously rebuilds calculus through limits, convergence, continuity, differentiability, integrals, and interchange of limit operations. Lecture notes and course texts freely available on MIT OCW.
Companion	Rudin — Principles of Mathematical Analysis	Standard reference (Baby Rudin). Dense but precise. Use alongside MIT notes for the formal definitions and theorem statements. The exercises are essential.
Discovery	awesome-math Analysis section —	For locating additional resources: Tao's Analysis I/II for a more pedagogical alternative to Rudin.

4.4 Mastery Test

Mastery Test

- ✓ Prove convergence of a sequence from the ε - N definition without appealing to intuition.
- ✓ Distinguish pointwise convergence from uniform convergence with a concrete counterexample.
- ✓ Prove that a continuous function on a closed bounded interval is uniformly continuous.
- ✓ State and apply the Banach fixed-point theorem, including the contraction condition.
- ✓ Explain why a limit-based argument works, not just that it feels correct.

5

PHA SE 3

Probability, Statistics, and Information Theory

Converting uncertainty from rhetoric into mathematics

5.1 Why This Phase Matters

You already reason in terms of evidence, uncertainty, entropy, gating, routing, and signal quality. This phase gives those ideas exact definitions and usable theorems. The information-theoretic track is particularly important: entropy, KL divergence, and mutual information appear throughout architecture design, training monitoring, and the theoretical foundations of the SCA spectral substrate.

5.2 Core Topics

- Probability axioms — Kolmogorov foundations, sample spaces, events, σ -algebras
- Random variables — discrete and continuous, cumulative distribution functions, density functions
- Expectation, variance, and covariance — linearity of expectation, variance decomposition
- Conditional probability and Bayes' theorem — conditional distributions, independence
- Concentration inequalities — Markov, Chebyshev, and Chernoff bounds
- Law of Large Numbers and Central Limit Theorem — convergence in probability and in distribution
- Statistical inference — maximum likelihood, Bayesian inference, hypothesis testing, confidence intervals
- Entropy and KL divergence — Shannon entropy, relative entropy, and their operational meanings
- Mutual information — as a measure of dependence, and its variational representations
- Source coding and channel capacity — data compression limits and the noisy channel theorem

5.3 Recommended Resources

Tier	Source	Description
Primary	MIT — Probability and Random Variables (6.041)	Distributions, conditional probability, Bayes' rule, joint distributions, variance, Chebyshev, LLN, CLT. MIT OCW, lecture notes and problem sets freely available.
Primary	MIT — Information Theory (6.441)	Entropy, divergence, mutual information, lossless compression, hypothesis testing, source coding, channel coding, and lossy compression. MIT OCW, lecture notes available.
Companion	MIT — Introduction to Probability and Statistics	Adds Bayesian inference, hypothesis testing, confidence intervals, and linear regression to the probability foundation.

Companion	OpenStax — Introductory Statistics 2e	Free peer-reviewed textbook. Use as a practical reference for statistical methods.
Advanced	Cover and Thomas — Elements of Information Theory	Standard graduate reference for information theory. Use alongside MIT notes once the basics are established.

5.4 Mastery Test

Mastery Test

- ✓ Compute $H(X)$, $H(X|Y)$, and $I(X;Y)$ for a small discrete joint distribution from scratch.
- ✓ Derive the KL divergence $D(P||Q)$ for two Gaussian distributions analytically.
- ✓ Apply Bayes' theorem to update a prior given two pieces of sequential evidence.
- ✓ State what it means for two random variables to be independent vs. uncorrelated.
- ✓ Explain the operational meaning of the entropy rate and the channel capacity theorem.

6

PHA SE 4

Differential Equations, Dynamical Systems, and Control

Regimes, attractors, stability, and feedback

6.1 Why This Phase Matters

This phase directly formalizes the native concepts that appear throughout the ToraFirma research programme: regimes, drift, attractors, bifurcation, stability, control surfaces, feedback, and hysteresis. The connection between ODEs and the Lax pair / Toda lattice framework is explicit, and Lyapunov theory provides the formal home for shell preservation arguments.

This phase will sharpen how you model training behavior, regime transitions, memory evolution, operator switching, and governed runtime dynamics. The MIT Underactuated Robotics notes are particularly valuable as they extend classical control theory to modern nonlinear and stochastic settings.

6.2 Core Topics

- **First- and second-order ODEs**
 - Separation of variables, integrating factors, characteristic equations.
 - Phase portraits for autonomous systems.
- **Linear systems and matrix exponentials**
 - $\dot{x} = Ax$: solution via matrix exponential and Jordan form.
 - Stability from eigenvalues of A .
- **Phase portraits and qualitative analysis**
 - Equilibria, nullclines, trajectories.
 - Classification of equilibria: stable/unstable nodes, saddles, centers, spirals.
- **Lyapunov stability theory**
 - Lyapunov functions as energy-like certificates.
 - LaSalle's invariance principle.
 - The discrete Lyapunov equation $A^T P A - P = -Q$ and shell preservation.
- **Bifurcations**
 - Saddle-node, pitchfork, Hopf bifurcations.
 - Parameter-dependent qualitative changes in dynamics.
- **State-space models and feedback**
 - Controllability and observability (Gramian rank conditions).
 - Pole placement, linear quadratic regulator (LQR) intuition.
- **Stochastic control and underactuated systems**
 - Exploiting natural dynamics rather than fighting them.
 - Stochastic differential equations and their stability analysis.

6.3 Recommended Resources

Tier	Source	Description
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Primary	MIT — Differential Equations (18.03)	ODEs as the language in which laws of nature are expressed. Lecture notes, videos, and independent-study materials. MIT OCW, freely available.
Primary	MIT — Underactuated Robotics	Exceptionally strong for modern control thinking, nonlinear dynamics, stochastic systems, and robust/stochastic control. Frames underactuated control as exploiting natural dynamics. Full lecture notes and videos available. Directly applicable to governed runtime architecture.
Companion	MIT — Learn Differential Equations: Up Close (Strang and Moler)	Computational viewpoint including Euler and Runge-Kutta methods. Good bridge between analytical and numerical treatment.
Companion	Stanford SEE — EE263	Linear dynamical systems from the control engineering perspective. Bridges Phase 1 and Phase 4.
Advanced	Strogatz — Nonlinear Dynamics and Chaos	Accessible treatment of phase portraits, bifurcations, and chaos. Strong geometric intuition. Use alongside the MIT materials for visual development.

6.4 Mastery Test

Mastery Test

- ✓ Linearize a nonlinear 2D system about an equilibrium and classify the equilibrium type.
- ✓ Draw a complete phase portrait for a 2D system with two equilibria.
- ✓ Construct a Lyapunov function for a simple nonlinear system and prove asymptotic stability.
- ✓ Explain what a Hopf bifurcation is and describe the qualitative behavior change it produces.
- ✓ State what the LQR problem is optimizing and what constraint the Riccati equation encodes.

7

PHA SE 5

Structure Modules: Algebra, Geometry, Complex Analysis

Symmetry, curvature, and structural depth

Phase 5 is less about immediate utility and more about deepening the structural vocabulary available for architectural reasoning. These three modules can be studied in parallel and are not strictly sequential with each other.

7.1 Abstract Algebra

Algebra sharpens the sense of composition, symmetry, lawful transformation, equivalence, homomorphism, and invariant structure. Groups, rings, and fields provide the precise language for describing when two systems are structurally identical versus merely similar.

- Groups and group actions — orbits, stabilizers, cosets, normal subgroups
- Rings and fields — ideals, quotient rings, polynomial rings
- Homomorphisms and isomorphisms — structure-preserving maps and their kernels
- Quotient structure — the first isomorphism theorem
- Symmetry groups — rotation groups, permutation groups, Cayley's theorem
- Representation theory — intuition for encoding group actions in linear maps

Tier	Source	Description
Primary	MIT — Algebra I (18.701)	Groups, vector spaces, linear transformations, symmetry groups, bilinear forms, and linear groups.
Primary	MIT — Modern Algebra (18.703)	Undergraduate algebra with strong application value in science and engineering.

7.2 Differential Geometry

Differential geometry provides rigorous foundations for reasoning about curvature, manifolds, metrics, and geometric constraints on state spaces. The distinction between local and global structure is a key conceptual output.

- Curves and surfaces — parametrization, tangent vectors, arc length
- Curvature — Frenet-Serret frame, Gaussian curvature, mean curvature
- Local vs global structure — the Gauss-Bonnet theorem as a prototype
- Geodesics and metric structure — shortest paths on curved surfaces
- Introduction to manifolds — tangent bundles, differential forms

Tier	Source	Description
Primary	MIT — Differential Geometry (18.950)	Rigorous treatment emphasizing concrete geometry centered on curvature. Lecture notes for plane curves, hypersurfaces, and lengths/distances. MIT OCW.

Companion	Do Carmo — Differential Geometry of Curves and Surfaces	Standard undergraduate reference. Excellent geometric intuition.
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7.3 Complex Analysis

Given the centrality of transforms, spectral reasoning, quadratures, phase, and operator structure in your work, complex analysis is not ornamental — it is structurally useful. The residue theorem connects directly to spectral integrals and the Abel map in the Sato Grassmannian framework.

- Analyticity and the Cauchy-Riemann equations
- Cauchy's integral theorem and formula
- Residues and the residue theorem — contour integration
- Laurent series and poles
- Harmonic functions and potential theory
- Conformal maps — geometric transformation under analytic functions
- Connections to Fourier and Laplace transforms

Tier	Source	Description
Primary	MIT — Complex Variables with Applications (18.04)	Analytic functions, Cauchy's theorem, contour integration, residues, Laurent series, harmonic functions, and potential theory. Lecture notes available. MIT OCW.
Primary	MIT — Functions of a Complex Variable (18.112)	More advanced treatment; covers the same core with greater depth.
Reference	Ahlfors — Complex Analysis	Standard graduate reference. Dense but authoritative.

8

PHA SE 6

Functional Analysis, Numerical Analysis, and Optimization

Infinite-dimensional structure, computation, and solvable formulations

8.1 Functional Analysis

Functional analysis is the formal home for the 'operator over operators' instinct that runs through your architectural work. Banach and Hilbert spaces generalize linear algebra to infinite-dimensional function spaces, and the spectral theorem for self-adjoint operators provides the rigorous foundation for the SCA spectral substrate.

- Normed spaces and Banach spaces — completeness, norms, examples
- Hilbert spaces — inner products, orthonormal bases, Gram-Schmidt
- Bounded linear operators — continuity, operator norms
- Hahn-Banach theorem — extension and separation
- Dual spaces — functionals, reflexivity, weak topologies
- Compact operators — finite-rank approximation, Fredholm theory
- Self-adjoint operators and the spectral theorem — functional calculus
- L^p spaces — measure theory background, Hölder and Minkowski inequalities

Tier	Source	Description
Primary	MIT — Introduction to Functional Analysis (18.102)	Normed spaces, completeness, Hahn-Banach, duality, operators, measure/ L^p spaces, Hilbert spaces, compact and self-adjoint operators, spectral theorem. Lecture notes and videos. MIT OCW.
Reference	Kreyszig — Introductory Functional Analysis with Applications	Accessible graduate reference with engineering motivation.
Reference	Reed and Simon — Methods of Modern Mathematical Physics, Vol. 1	Authoritative treatment for those moving into mathematical physics applications.

8.2 Numerical Analysis

Numerical analysis is critical for ensuring that mathematical systems survive contact with computation. Conditioning, stability, and floating-point effects are the fault lines where elegant theory meets engineering reality.

- Floating-point arithmetic — machine epsilon, catastrophic cancellation, IEEE 754
- Conditioning and stability — condition numbers, backward and forward error analysis
- Root finding — Newton's method, bisection, convergence rates

- Interpolation and approximation — polynomial, spline, Chebyshev
- Numerical integration — Gaussian quadrature, adaptive methods, error bounds
- Numerical linear algebra — LU, QR, Cholesky, iterative methods (CG, GMRES)
- Numerical ODEs — Euler, Runge-Kutta, stiffness, stability regions
- Sparse methods and iterative solvers for large-scale systems

Tier	Source	Description
Primary	MIT — Introduction to Numerical Analysis (18.330)	Root finding, interpolation, approximation, numerical integration, differential equations, floating-point arithmetic, conditioning, stability. MIT OCW.
Primary	MIT — Introduction to Numerical Methods (18.335)	Sparse/dense linear algebra, iterative methods, and more advanced numerical techniques.
Reference	Trefethen and Bau — Numerical Linear Algebra	Best reference for the linear algebra side; geometrically motivated.

8.3 Optimization

Optimization is where many abstract mathematical ideas become concrete decision procedures.

Convexity is the key structural property that separates tractable from intractable. Duality theory provides complementary perspectives on every optimization problem.

- Convex sets and convex functions — definitions, operations preserving convexity
- Unconstrained optimization — gradient descent, Newton's method, convergence rates
- Constrained optimization — Lagrangians, KKT conditions
- Duality — weak and strong duality, Slater's condition, Lagrangian duality
- Semidefinite programming — LMI formulations, connection to Lyapunov analysis
- Dynamic programming — Bellman equations, value functions, HJB equation
- Optimal control — LQR as a constrained optimization problem, Pontryagin principle
- Robust optimization — worst-case formulations, min-max problems

Tier	Source	Description
Primary	Stanford — Convex Optimization (Boyd and Vandenberghe)	Lecture slides, notebooks, videos, and the standard convex optimization text. Free materials and textbook available from Stanford.
Primary	MIT — Optimization (6.079 / 15.093)	Covers linear, discrete, nonlinear, dynamic optimization, robust optimization, and optimal control.
Reference	Boyd and Vandenberghe — Convex Optimization	The standard reference text. Freely available as a PDF from Stanford.

Synthesis Track — Mathematics and Architecture

The synthesis track is what most curricula omit. It is the mechanism by which formal mathematics becomes architectural capability. It runs in parallel throughout the entire programme — not as a separate phase to complete at the end.

9.1 Translation Notebook Protocol

For every major concept encountered across all phases, record a structured entry in a translation notebook with the following five fields:

Formal object	The precise mathematical definition and its notation.
Theorem or law	The key result that makes this object useful.
Intuitive meaning	What this object does geometrically or operationally.
System interpretation	Where and how this appears in architectural work.
Failure mode	What breaks when the assumptions of the theorem fail.

9.2 Translation Table: Formal Objects to System Design

The following table provides example entries illustrating the translation discipline. The architectural interpretations are written for the specific context of spectral architecture, governed AI inference, and training dynamics.

Concept	Eigenvalue
Theorem / Law	If $Ax = \lambda x$, then λ is an eigenvalue. Stability: discrete system stable iff $ \lambda < 1$ for all eigenvalues.
Meaning	A growth or decay mode of a linear system.
System Use	Koopman eigenvalues encode regime persistence. Training stability governed by spectral radius of the Jacobian.
Failure Mode	Eigenvalues of the full nonlinear Jacobian need not predict nonlinear stability at large deviations.

Concept	Projection
Theorem / Law	$P^2 = P$ (idempotent). The projection of x onto subspace V minimizes $\ x - Px\ $.
Meaning	Admissible reduction: the best approximation of x within V .
System Use	KV cache eviction via PageRank is an approximate projection onto the most informative subspace.

Failure Mode	Projection does not preserve norm unless the projection is orthogonal.
Concept	Shannon Entropy
Theorem / Law	$H(X) = -\sum p(x) \log p(x)$. Maximized by the uniform distribution.
Meaning	The average uncertainty in a random variable X .
System Use	Entropy-based routing gates: high entropy signals regime uncertainty requiring spectral intervention.
Failure Mode	Entropy measures uncertainty, not importance. A deterministic but hard problem has zero entropy.
Concept	Lyapunov Function
Theorem / Law	$V(x) > 0$, $V(0) = 0$, $dV/dt < 0$ along trajectories implies asymptotic stability.
Meaning	A generalized energy function that decreases along trajectories.
System Use	Shell preservation: $V_t = \mathbf{J}_t - \mathbf{J}^* \mathbf{F}$, if it satisfies the Lyapunov conditions, certifies spectral stability of the governed transformer.
Failure Mode	V_t is not automatically a Lyapunov function — it must satisfy the decay condition explicitly. This is the V_t fallacy: load-measuring valence fields are NOT Lyapunov functions.
Concept	Compact Operator
Theorem / Law	T is compact if it maps bounded sets to precompact sets (relatively compact closure).
Meaning	A tractable approximation regime — the operator behaves 'almost finite-dimensional'.
System Use	Fisher information operators in FILQ quantization: compactness ensures finite-rank approximability.
Failure Mode	Non-compact operators (e.g., unbounded multiplication operators) do not admit the same spectral theory.
Concept	Curvature
Theorem / Law	$\kappa = dT/ds$ (rate of change of unit tangent). Gaussian curvature $K = \kappa_1 \kappa_2$.
Meaning	Local deformation of a curve or surface relative to flat space.
System Use	Curvature of the loss landscape: high curvature $\rightarrow$ poor conditioning $\rightarrow$ sensitivity to learning rate.
Failure Mode	Curvature is a local quantity — flat curvature locally does not imply global convexity.

9.3 Mathematical Artifact Habit

Every month, produce one mathematical artifact. The artifact must be written, not merely understood. Acceptable formats:

- **Proof memo** — a self-contained proof of one theorem encountered during study
- **Derivation memo** — a step-by-step derivation connecting two ideas
- **Theorem-to-system translation** — a formal mapping from a mathematical result to an architectural component
- **Simulation notebook** — code + analysis demonstrating a mathematical phenomenon computationally
- **Counterexample notebook** — examples showing where a theorem fails when assumptions are violated
- **'Where the analogy breaks' memo** — explicit identification of the boundary of a structural isomorphism

Why 'Where the Analogy Breaks' Matters

- Analogical reasoning is a primary discovery tool but a dangerous load-bearing tool.
- Most architectural errors do not come from bad intuitions — they come from overextended analogies.
- The 'where it breaks' memo forces explicit identification of the point at which a structural similarity is merely similarity rather than isomorphism.
- This directly implements the second capability goal: separate analogy from actual structural equivalence.

Study Cadence and Three-Pass Method

10.1 Weekly Session Structure

Each week should contain four sessions. These sessions are not optional components — the cadence keeps material from becoming either purely symbolic or purely intuitive.

Session A Theory	Read definitions, theorems, and worked examples. Focus on understanding the objects and what the theorem is asserting. Do not skip formal definitions in favor of intuitive summaries.
Session B Proof	Do proof exercises and derivation drills. If stuck for more than 15 minutes, record the failure in the error log and look at the solution — then re-prove it independently.
Session C Computation	Implement a small numerical experiment: solve an ODE, compute an SVD, estimate entropy, simulate a Markov chain, test conditioning, or optimize a convex objective.
Session D Translation	Write one page connecting the mathematics studied this week to your architectures. Produce one translation notebook entry. If a theme is not yet connectable, note why.

10.2 Three-Pass Method

For your profile, each topic should be learned in three passes rather than a single textbook-linear reading. The three-pass method prevents the common failure mode of building intuition without precision, or precision without engineering utility.

Pass 1 — Intuition

- What objects exist in this theory?
- What do they mean geometrically or operationally?
- What problems does this theory solve?
- Sketch the key objects without worrying about precise definitions.

Pass 2 — Formal Layer

- Write out definitions precisely.
- State the key theorems with their hypotheses.
- Understand at least one proof for each major theorem.
- Identify the load-bearing assumptions.

Pass 3 — Engineering Layer

- What breaks if the assumptions fail?
- How does this appear computationally?
- What can be measured or tested?
- Write one translation notebook entry.

Resource Policy and Hierarchy

Use resources in strict tier order. The primary spine provides pedagogically ordered materials with full problem sets and proofs. Secondary discovery sources are curated indexes for finding additional references but must not define the sequence. Tertiary materials should only be engaged once the primary spine is stable.

Tier	Source	Description
P1	MIT OpenCourseWare (OCW)	Official lecture notes, problem sets, exams, and videos for all core phases. Free, no registration. Primary source for Phases 0–4 and Phase 6.
P1	Stanford Engineering Everywhere (SEE)	Full course materials for EE263 (linear dynamical systems) and related courses. Bridges linear algebra and control theory.
P1	MIT Underactuated Robotics Notes	Russ Tedrake's lecture notes and videos. Primary source for nonlinear dynamics and modern control in Phase 4. Current working material with ongoing updates.
P1	OpenStax	Free peer-reviewed textbooks for calculus (Phase 0) and statistics (Phase 3). No-cost, openly licensed.
P1	Stanford — Convex Optimization (Boyd)	Lecture slides, textbook PDF, and course notes. Freely available from Stanford. Primary source for the optimization module in Phase 6.
S2	awesome-math	Curated GitHub index of mathematics resources. Use for discovery only — do not let it define the sequence. Useful for locating specialist books and notes.
S2	awesome (general)	Broader curated discovery index. Use as a secondary search layer for finding domain-specific mathematical tools adjacent to your research areas.
T3	Research papers	Domain-specific mathematical papers relevant to SCA, Toda lattice, Sato Grassmannian, and integrable systems. Engage once Phase 5/6 spine is stable.
T3	Specialist lecture notes	Advanced notes not covered by MIT OCW: e.g., Terry Tao's blog posts, Villani's optimal transport notes, or Lax's functional analysis lectures.

Recommended Emphasis Order

The following order maximizes return relative to your current research interests: representational power first, then rigor, then dynamics and control intuition, then operator-theoretic depth.

#	Phase	Subject	Priority Rationale
1	Phase 0	Proofs and Discrete Math	Non-negotiable foundation. Short phase if proof discipline is already present.
2	Phase 1	Linear Algebra	Highest return per unit time. Directly sharpens operator and representation intuitions.
3	Phase 2	Real Analysis	Hardens the informal convergence and stability claims that appear in every other phase.
4	Phase 3	Probability + Information Theory	Formalizes entropy, uncertainty, and inference language used throughout architectural work.
5	Phase 4	Differential Equations + Control	Directly formalizes training dynamics, regime transitions, and governed runtime.
6	Phase 5a	Complex Analysis	Structurally useful for spectral integrals, transforms, and Abel map geometry.
7	Phase 5b	Abstract Algebra	Deepens structural vocabulary; supports isomorphism reasoning and symmetry analysis.
8	Phase 5c	Differential Geometry	Provides the geometric language for manifold-valued parameters and curvature reasoning.
9	Phase 6a	Functional Analysis	Formal home for operator-over-operator reasoning and infinite-dimensional spectral theory.
10	Phase 6b	Numerical Analysis	Ensures mathematical systems survive contact with computation.
11	Phase 6c	Optimization	Converts abstract optimization ideas into tractable decision procedures.

Mastery Checkpoints

These checkpoints serve as gates — not as encouragement milestones. Each checkpoint should be verified by producing a written artifact that demonstrates the capability, not merely by a sense of understanding.

Checkpoint	Mastery Criterion
CP-1	Write clean proofs and manipulate discrete structures without ambiguity. Specifically: produce a proof by induction and a proof by contradiction, both reviewed for logical gaps.
CP-2	Think in operators, eigenspaces, projections, and linear state-space models. Specifically: given a matrix, characterize its four fundamental subspaces, compute its SVD, and explain its eigenstructure and stability properties.
CP-3	Distinguish convergence claims rigorously and know when a limit argument is invalid. Specifically: produce a counterexample to a false interchange of limits and prove a convergence claim from the epsilon-N definition.
CP-4	Speak precisely about uncertainty, entropy, inference, and evidence quality. Specifically: compute H , $H(X Y)$, $I(X;Y)$, and $D(P Q)$ for a non-trivial joint distribution; derive a Bayesian posterior update from scratch.
CP-5	Model simple dynamical systems, reason about stability, and state what feedback is doing. Specifically: construct a phase portrait, find equilibria, classify them, and build a Lyapunov function for one stable equilibrium.
CP-6	Recognize when a structural idea is algebraic, geometric, analytic, probabilistic, or numerical in nature. Specifically: given a new concept, correctly classify it and identify which phase provides its formal home.
CP-7	Translate a new architecture idea into objects, assumptions, invariants, dynamics, observables, and possible proofs or failure modes. This is the terminal checkpoint. Specifically: take a current architectural component and produce a complete translation notebook entry with all five fields filled with precise mathematical content.

Note on CP-7

- CP-7 is the real target of the entire curriculum.
- It is satisfied not when you can describe a mathematical concept, but when you can use it to decompose an architectural claim into load-bearing components that can each be individually proved, falsified, or measured.
- The proof programme methodology (Class 1/2/3 evidence hierarchy, NT/ET/F ledger) is the natural output format for CP-7 products.

Final Operational Rule

**Build your mathematics around proof, operators, dynamics,
uncertainty, and computation.
Maintain a live translation layer back into your architectures.**

This rule encodes the productive tension that runs throughout the curriculum: mathematical depth is not useful in isolation, and architectural intuition is not trustworthy without formal backing. The two sides must be maintained simultaneously via the synthesis track.

This rule will give you the strongest gain in:

- **Rigor** — the ability to know when a claim is proved versus merely plausible
- **Expressive power** — the ability to state ideas precisely enough to be falsifiable
- **Falsification ability** — the ability to identify and test load-bearing assumptions
- **Architectural compression** — the ability to compress a complex system into its essential structure

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Appendix: Complete Resource List

A.1 Online Courses and Lecture Notes (Free)

ID	Title	URL
MIT OCW 18.062J	Mathematics for Computer Science	https://ocw.mit.edu/courses/6-042j-mathematics-for-computer-science/
MIT OCW 18.06	Linear Algebra (Gilbert Strang)	https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/
MIT OCW 18.06SC	Linear Algebra (Self-Study)	https://ocw.mit.edu/courses/18-06sc-linear-algebra-fall-2011/
MIT OCW	Matrix Calculus for ML and Beyond	https://ocw.mit.edu/courses/18-s096-matrix-calculus-for-machine-learning-and-beyond-january-iap-2023/
Stanford SEE	EE263: Intro to Linear Dynamical Systems	https://see.stanford.edu/Course/EE263
MIT OCW 18.100	Real Analysis	https://ocw.mit.edu/courses/18-100a-real-analysis-fall-2020/
MIT OCW 6.041	Probability and Random Variables	https://ocw.mit.edu/courses/6-041-probabilistic-systems-analysis-and-applied-probability-fall-2010/
MIT OCW 6.441	Information Theory	https://ocw.mit.edu/courses/6-441-information-theory-spring-2016/
MIT OCW 18.03	Differential Equations	https://ocw.mit.edu/courses/18-03-differential-equations-spring-2010/
MIT	Underactuated Robotics	https://underactuated.mit.edu/
MIT OCW 18.701	Algebra I	https://ocw.mit.edu/courses/18-701-algebra-i-fall-2010/
MIT OCW 18.950	Differential Geometry	https://ocw.mit.edu/courses/18-950-differential-geometry-fall-2008/
MIT OCW 18.04	Complex Variables with Applications	https://ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/
MIT OCW 18.102	Introduction to Functional Analysis	https://ocw.mit.edu/courses/18-102-introduction-to-functional-analysis-spring-2021/
MIT OCW 18.330	Introduction to Numerical Analysis	https://ocw.mit.edu/courses/18-330-introduction-to-numerical-analysis-spring-2012/

Stanford	Convex Optimization (Boyd and Vandenberghe)	https://web.stanford.edu/~boyd/cvxbook/
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A.2 Free Textbooks

Title	Notes	URL
OpenStax Calculus Vol 1–3	Full calculus sequence, peer-reviewed	https://openstax.org/subjects/math
OpenStax Introductory Statistics 2e	Statistics and probability	https://openstax.org/books/introductory-statistics-2e/
Boyd and Vandenberghe — Convex Optimization	Full textbook PDF	https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf
Tao — Analysis I and II	Rigorous real analysis (available via author's website)	https://terrytao.wordpress.com/books/
Axler — Linear Algebra Done Right (Open Access)	Coordinate-free treatment	https://linear.axler.net/

A.3 Reference Books (Not Free)

Author(s)	Title	Notes
Rudin	<i>Principles of Mathematical Analysis</i>	Standard real analysis reference. Dense, precise, essential exercises.
Horn and Johnson	<i>Matrix Analysis</i>	Authoritative reference for advanced linear algebra and matrix theory.
Strogatz	<i>Nonlinear Dynamics and Chaos</i>	Best accessible treatment of phase portraits, bifurcations, and chaos.
Cover and Thomas	<i>Elements of Information Theory</i>	Standard graduate reference for information theory.
Kreyszig	<i>Introductory Functional Analysis with Applications</i>	Accessible functional analysis with engineering motivation.
Do Carmo	<i>Differential Geometry of Curves and Surfaces</i>	Standard undergraduate differential geometry reference.
Ahlfors	<i>Complex Analysis</i>	Authoritative complex analysis graduate reference.
Trefethen and Bau	<i>Numerical Linear Algebra</i>	Best reference for numerical linear algebra; geometrically motivated.

Reed and Simon	<i>Methods of Modern Mathematical Physics Vol. 1</i>	Authoritative functional analysis for mathematical physics applications.
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A.4 Discovery Indices

- **awesome-math** (github.com/rossant/awesome-math) — curated list of mathematics resources organized by subject
- **awesome** (github.com/sindresorhus/awesome) — broader curated resource index; use math subcategory
- **Terry Tao's Blog** (terrytao.wordpress.com) — research commentary, lecture notes, and expository articles
- **nLab** (ncatlab.org) — category-theoretic and structural mathematics reference; use selectively
- **arXiv math-ph** / **math.CA** / **math.SP** — primary source for advanced integrable systems and spectral theory